Ex-31 - Action on a vector field

1) $$g\cdot f = \left(\frac{dg}{dx}\cdot f\right)\circ g^{-1}$$

$$g\cdot f(y) = \left(\frac{dg}{dx} * f\right)\circ g^{-1}(y)$$

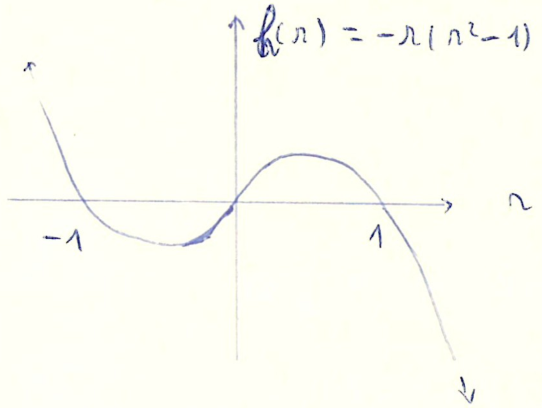
$$\overset{\text{if } y=g(x)}{=} \left(\frac{dg}{dx}\cdot f\right)\circ g^{-1}(g(x))$$

$$= \left(\frac{dg}{dx}\cdot f\right)\circ(x)$$

$= \frac{dg}{dx}\cdot f(x)$ but $f(x)=\dot{x}$ so $g\cdot f(y) = \frac{dg}{dx}\cdot\frac{dx}{dt} = \frac{d(g(x))}{dt} = \frac{d(y)}{dt}$

$$\boxed{\text{So if } y=g(x),\ g\cdot f(y)=\dot{y}}$$

2)



If $\dot{r} = h(r)$, then:

if $r\in[0,1]$, $h(r)\geq 0 \Rightarrow \dot{r}\geq 0 \Rightarrow r\nearrow 1^-$

if $r\in[1,+\infty[$, $h(r)\leq 0 \Rightarrow \dot{r}\leq 0 \Rightarrow r\searrow 1^+$

So r will converge to 1 when $t\rightarrow+\infty$

3) We have $x = g(r,\theta) = r\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix}$

So $\dot{x} = \dot{r}\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix} + r\begin{pmatrix}-\dot{\theta}\sin\theta \\ \dot{\theta}\cos\theta\end{pmatrix}$

$$= \begin{pmatrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{pmatrix}\begin{pmatrix}\dot{r} \\ \dot{\theta}\end{pmatrix} \quad \text{but} \begin{pmatrix}\dot{r} \\ \dot{\theta}\end{pmatrix} = \begin{pmatrix}-r(r^2-1) \\ 1\end{pmatrix}$$

So $\dot{x} = \begin{pmatrix}\cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta\end{pmatrix}\begin{pmatrix}-r(r^2-1) \\ 1\end{pmatrix} = \begin{pmatrix}-r(r^2-1)\cos\theta - r\sin\theta \\ -r(r^2-1)\sin\theta + r\cos\theta\end{pmatrix}$

but $r = \|x\| = \sqrt{x_1^2+x_2^2}$ and $\cos\theta = \frac{x_1}{\|x\|}$ and $\sin\theta = \frac{x_2}{\|x\|}$

So: $\dot{x} = \begin{pmatrix}-x_1(x_1^2+x_2^2-1) - x_2 \\ -x_2(x_1^2+x_2^2-1) + x_1\end{pmatrix}$

$$\boxed{\dot{x} = \begin{pmatrix}-x_1^3 - x_1x_2^2 + x_1 - x_2 \\ -x_2^3 - x_1^2x_2 + x_1 + x_2\end{pmatrix}}$$

4) We have $g(x) = Ax$

We call $y = g(x)$. So $\dot{y} = \frac{d}{dt}(y) = A\dot{x} = A \cdot f(x)$

But from 1), we have if $y = g(x)$, then $\dot{y} = g \circ f(y)$

So $A f(x) = g \circ f(Ax)$. $\quad = g \circ f(Ax)$

Let's use $X = A^{-1} \cdot x$, then $A f(X) = g \circ f(AX) \iff \boxed{A f A^{-1} \cdot x = (g \circ f)(x)}$

5) We will use a matrix A composed of different transformation:

$A = R \cdot E \cdot S$ where $S = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}$ a symetry for the clockwise

$E = \begin{pmatrix}2 & 0 \\ 0 & 1\end{pmatrix}$ an ellipse kind of matrix

$R = \begin{pmatrix}\cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4}\end{pmatrix}$ a rotation matrix.